Guidelines for reviewing chemistry questions

Check whether each question is clear and useful, and whether the document supports what the question assumes and what the answer says. Score each check separately; one problem need not lower every score.

Read the example document below before reviewing the pairs. Some wording has been adapted to show a particular problem. "Good" and "Bad" apply only to the check being illustrated; a pair can do well on one check and poorly on another. Each suggested score and reason concern only that check.

Score	Meaning
5 — Exceptional	Fully successful, with a specific strength beyond ordinary good execution. Explain it.
4 — Fully successful	Strong, complete and correct for this criterion; no flaw is necessary.
3 — Limited	A specific minor limitation, or uncertainty that the supplied text cannot resolve. Name it.
2 — Significant problem	An important problem requires correction.
1 — Fails	The essential requirement is not met, or the essential information is missing.

Use **cannot assess**, with a reason, only when missing material or expertise prevents judgment. A known unsupported answer is a failure, not an unassessable case.

Example document

Complete source text:

The invention provides a paint spraying method for a plastic shell of an electric bicycle. The paint spraying method comprises the steps that (1) degreasing is conducted, specifically, spraying and cleaning are conducted with an aqueous degreasing agent containing alkali liquor and a surface active agent, and oil dirt on the surface of plastic is removed through the saponification of alkali and animal and vegetable oil in a solution and the emulsion process of the surface active agent and mineral oil; degreasing comprises a pre-degreasing procedure and a degreasing procedure, the pre-degreasing procedure is conducted for 1-2 min, and the degreasing procedure is conducted for 2-3 min; and the spraying pressure is 0.1-0.2 MPa; and the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent; and (b) interface activation is conducted, specifically, treatment is conducted through an aqueous surface adjusting agent containing aqueous macromolecule, an indoor-temperature spraying is conducted for 1 min, and the spraying pressure is 0.05-0.1 MPa. The paint spraying method is applied to paint spraying of the plastic shell of the electric bicycle.

1. Are the question and answer supported by the text? (Grounding)

Grounding means that the document supports the answer and the factual assumptions in the question. Check chemical identities, conditions and the distinction between proposed uses and demonstrated effects. An exact quotation may answer the wrong question. Give at most 2 for an important unsupported statement. A statement presented for checking need not be true.

Good — **Question:** "What is the composition of the aqueous degreasing agent used in the degreasing step of the paint spraying method for a plastic shell of an electric bicycle?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 4/5.** All four components and their concentration ranges are directly supported by the document.

Bad — **Question:** "What is the composition of the aqueous degreasing agent used in the degreasing step of the paint spraying method for a plastic shell of an electric bicycle?" **Answer:** "The agent contains sodium hydroxide, sodium phosphate, sodium carbonate, a surface active agent and zinc oxide." **Score: 2/5.** Zinc oxide is added to the answer without any support in the document.

2. Does the answer include the right amount of evidence? (Precision)

Precision means including the evidence needed to answer the question, with essential qualifications and no unnecessary material. Deduct for excess only if a shorter continuous extract suffices. Give 3 for minor excess and at most 2 for a decisive omission.

Good — **Question:** "What is the composition of the aqueous degreasing agent used in the degreasing step of the paint spraying method for a plastic shell of an electric bicycle?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 4/5.** All four requested components and their concentrations are included without unnecessary surrounding text.

Bad — **Question:** "What concentration range of sodium hydroxide is specified for the aqueous degreasing agent used in degreasing a plastic electric bicycle shell before painting?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 3/5.** Only sodium hydroxide is asked for; the other three components are unnecessary.

3. Are numbers and references accurate? (Numerical fidelity)

Numerical fidelity means preserving numbers and references accurately, including their meaning in context. Check values, units, ranges, substances, stages and percentage bases. Give at most 2 for an important change. If inapplicable, enter 4 and "not applicable." Assess missing requested numbers under checks 1 and 2.

Good — **Question:** "What concentration range of sodium hydroxide is specified for the aqueous degreasing agent used in degreasing a plastic electric bicycle shell before painting?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 4/5.** Each concentration retains the correct range, unit and corresponding component from the document.

Bad — **Question:** "What concentration range of sodium hydroxide is specified for the aqueous degreasing agent used in degreasing a plastic electric bicycle shell before painting?" **Answer:** "the aqueous degreasing agent is composed of 50-70 mg/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 1/5.** Changing g/L to mg/L makes the requested concentration 1,000 times too low.

4. Would an expert reasonably need this information? (Search realism)

Search realism is the plausibility of a question as a real request for information. Consider whether a researcher or practitioner would seek it. Simple questions can succeed fully. A question stating its complete answer seeks nothing new. Judge this separately from the answer's success.

Good — **Question:** "In the paint spraying method for an electric bicycle plastic shell, what are the pre-degreasing time, the degreasing time, and the spraying pressure during degreasing?" **Answer:** "degreasing comprises a pre-degreasing procedure and a degreasing procedure, the pre-degreasing procedure is conducted for 1-2 min, and the degreasing procedure is conducted for 2-3 min; and the spraying pressure is 0.1-0.2 MPa;" **Score: 4/5.** The requested operating times and pressure are useful details for investigating the process.

Bad — **Question:** "In paint spraying for electric bicycle plastic shells, the pre-degreasing procedure lasts 1-2 min. How long does pre-degreasing last?" **Answer:** "1-2 min" **Score: 1/5.** The question already states the duration, so there is nothing to find out.

5. Is the question understandable on its own? (Standalone clarity)

Standalone clarity means the question can be understood without seeing the source document. Check that it identifies the subject, conditions and information sought without relying on an unseen formula, drawing or passage. A named chemical class may suffice. Give at most 2 for essential missing context; 1 for an unidentifiable subject.

Good — **Question:** "What is the specified composition of the aqueous degreasing agent used when spraying paint onto a plastic shell of an electric bicycle?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent;" **Score: 4/5.** The question identifies the agent, its use and the relevant product.

Bad — **Question:** "What is its composition?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent;" **Score: 1/5.** "Its" gives no way to identify the material without reading the document.

6. Does the answer provide the information asked for? (Information value)

Information value is the extent to which the answer provides what the question asks for. Check every requested detail and necessary condition for the same subject; related facts alone are insufficient. Give at most 2 if an important part is missing; 1 if the main question is unanswered. Do not demand explanation beyond what answers the request.

Good — Question: "For the pre-degreasing and degreasing procedures in the degreasing step of the paint spraying method for a plastic shell of an electric bicycle, what duration is specified for each procedure and what spraying pressure is used?" **Answer:** "the pre-degreasing procedure is conducted for 1-2 min, and the degreasing procedure is conducted for 2-3 min; and the spraying pressure is 0.1-0.2 MPa" **Score: 4/5.** Both requested durations and the spraying pressure are present, resolving the whole request.

Bad — Question: "For the pre-degreasing and degreasing procedures in the degreasing step of the paint spraying method for a plastic shell of an electric bicycle, what duration is specified for each procedure and what spraying pressure is used?" **Answer:** "the pre-degreasing procedure is conducted for 1-2 min" **Score: 2/5.** The answer omits the degreasing duration and spraying pressure, both explicitly requested.

7. Does the question make one clear, focused request? (Focus and formulation)

Focus and formulation describe how clearly a question expresses one purpose, using only the wording needed. Allow connected details in one request. Identify specific repetition or unrelated demands before deducting. Judge the question's wording, not the answer's length. Keep necessary chemical names.

Good — Question: "What concentration range of sodium hydroxide is specified for the aqueous degreasing agent used in degreasing a plastic electric bicycle shell before painting?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 4/5.** The question asks for one concentration range and gives the necessary process context.

Bad — Question: "For the plastic shell of an electric bicycle, meaning an electric bicycle's plastic shell, what concentration range of sodium hydroxide is specified for the aqueous degreasing agent?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 3/5.** Repeating the plastic-shell description adds words without clarifying the request.

8. Is the language clear and natural? (Linguistic quality)

Linguistic quality is the correctness and naturalness of the question's language. Assess grammar, terminology and sentence structure in its original language. Give 4 for clear ordinary writing, 3 for a specific awkward expression, 2 for language that impedes reading, or 1 if unintelligible. Do not deduct solely for an unsupported answer or copied language errors.

Good — Question: "What concentration range of sodium hydroxide is specified for the aqueous degreasing agent used in degreasing a plastic electric bicycle shell before painting?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 4/5.** The question is grammatical, natural and clear, with appropriate chemical terminology.

Bad — Question: "For paint spray electric bicycle plastic shell, what the sodium hydroxide concentration aqueous degreasing agent are?" **Answer:** "the aqueous degreasing agent is composed of 50-70 g/L of sodium hydroxide, 20-30 g/L of sodium phosphate, 10-20 g/L of sodium carbonate, and 5-10 g/L of the surface active agent" **Score: 2/5.** Broken word order and verb agreement make the question difficult to read.

Record all eight scores with brief reasons and any needed corrections.

Qualification test: understanding the task

Annotator ID: _____ Date: _____

Choose **exactly one answer (A–D)** for each question and circle your choice. You may consult this guide; work independently. Submit the test for review before beginning annotation.

1. For an item you can assess, what must you record?

A. One overall 1–5 score; **B.** Eight scores without reasons; **C.** Eight 1–5 scores, reasons and needed corrections; **D.** A rewritten question and answer only.

2. How should you distinguish scores 4 and 5?

A. 4: fully successful; 5: an explained exceptional strength; **B.** 4: flawed; 5: error-free; **C.** 4: simple; 5: difficult; **D.** 4 and 5 mean the same thing.

3. How should an important unsupported assertion affect scoring?

A. Give every criterion 1; **B.** Give Grounding 3 and leave other scores unchanged; **C.** Mark every criterion "cannot assess"; **D.** Give Grounding at most 2; judge other criteria separately.

4. No relevant numbers or references apply. What do you record for Numerical fidelity?

A. 1; **B.** 4 and "not applicable"; **C.** 5; **D.** "Cannot assess".

5. When should you use "cannot assess"?

A. The answer is known to be unsupported; **B.** The wording is poor; **C.** Any score would be below 4; **D.** Needed material or expertise is missing; explain why.

6. What do you use to judge factual support and Linguistic quality, respectively?

A. Supplied text; the question's original language; **B.** Supplied text; an English translation only; **C.** Outside sources; the original language; **D.** Outside sources; an English translation only.

7. When checking whether a question stands on its own, how should you handle missing context?

A. Fill it in from the document; **B.** Assume the most likely subject; **C.** Judge only the information in the question; **D.** Ignore missing context if the grammar is correct.

8. Before lowering a score because an answer is too long, what should you check?

A. Whether it exceeds a fixed word limit; **B.** Whether a shorter continuous passage preserves all needed information and qualifications; **C.** Whether another answer is shorter; **D.** Whether technical terms can be removed.

9. What should a useful explanation of a score include?

A. The score number repeated in words; **B.** A general impression such as "good" or "bad"; **C.** A specific strength or problem relevant to that criterion; **D.** A comparison with the previous item.

10. What should you do when an answer provides only part of the requested information?

A. Give full marks because the supplied facts are correct; **B.** Identify what is missing and lower the affected scores appropriately; **C.** Give every criterion 1; **D.** Use "cannot assess" for every criterion.